LEDU_exercise3

Latent dirichlet allocation

```
# Copy-pasted code from the exercise's page
library(pacman)
library(remotes)
library(text2map.corpora)
library(reshape2)
library(dplyr)
```

```
Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

    filter, lag

The following objects are masked from 'package:base':

    intersect, setdiff, setequal, union
```

```
pacman::p_load(tidyverse,
               textclean,
               textstem,
               stringi,
               tidytext,
               text2map,
               topicmodels,
               topicdoc,
               Matrix,
```

```
               forcats,
               LDAvis,
               servr,
               devtools,
               quanteda,
               udpipe,
               textclean,
               stm,
               install = T)

devtools::install_github("quanteda/quanteda.corpora")
```

```
Warning: `install_github()` was deprecated in devtools 2.5.0.
i Please use pak::pak("user/repo") instead.
```

```
Skipping install of 'quanteda.corpora' from a github remote, the SHA1 (5259da4d) has not cha
  Use `force = TRUE` to force installation
```

```
library(quanteda.corpora)

data("data_corpus_ungd2017", package = "quanteda.corpora")
un <- data_corpus_ungd2017
rm(data_corpus_ungd2017) #rename the file to "un" to make the script cleaner

un <- un |>
  convert(to = "data.frame") |>
  drop_na(continent, gdp_per_capita)
```

```
# Code copy-pasted from the exercise's page
un <- un |>
  mutate(text.clean = stri_trans_general(text, id = "Any-Latin;Latin-ASCII"),
         text.clean = tolower(text.clean),
         text.clean = replace_curly_quote(text.clean),
         text.clean = replace_contraction(text.clean),
         text.clean = gsub("[[:digit:]]+", " ", text.clean),
         text.clean = gsub("[[:space:]]+", " ", text.clean),
         text.clean = trimws(text.clean))
```

```
# Code copy-pasted from the exercise's page
eng <- udpipe_download_model(language = "english-gum")
```

```
Downloading udpipe model from https://raw.githubusercontent.com/jwijffels/udpipe.models.ud.2
ud-2.5-191206/english-gum-ud-2.5-191206.udpipe to /home/onyxia/work/css-
class/Lecture 3/english-gum-ud-2.5-191206.udpipe

 - This model has been trained on version 2.5 of data from https://universaldependencies.org

 - The model is distributed under the CC-BY-SA-NC license: https://creativecommons.org/licens
nc-sa/4.0

 - Visit https://github.com/jwijffels/udpipe.models.ud.2.5 for model license details.

 - For a list of all models and their licenses (most models you can download with this packag
BY-SA or a CC-BY-SA-NC license) read the documentation at ?udpipe_download_model. For buildi
train', package = 'udpipe')

Downloading finished, model stored at '/home/onyxia/work/css-class/Lecture 3/english-
gum-ud-2.5-191206.udpipe'
```

```r
ud_eng <- udpipe_load_model(eng$file_model)

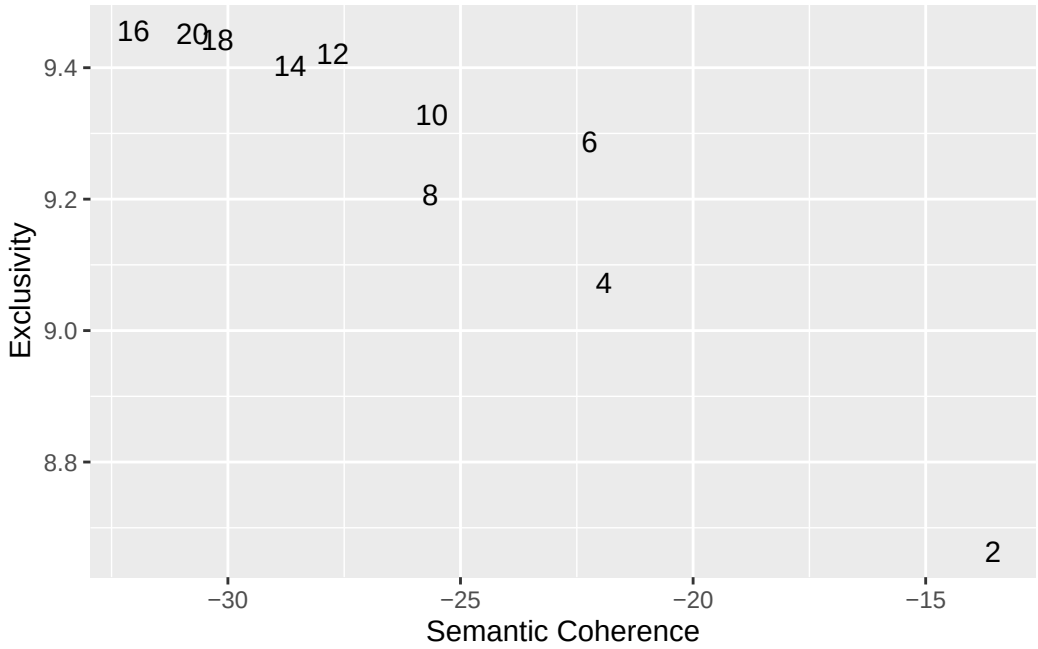
un.df.trip <- udpipe_annotate(
  x = un$text.clean,
  object = ud_eng) |>
  as_tibble() |>
  anti_join(stop_words, by = c("lemma" = "word")) |>
  select(doc_id, lemma) |>
  filter(!grepl("[[:punct:]]+", lemma)) |>
  count(doc_id, lemma)
```

```r
# Making the document term matrix
un.dtm <- un.df.trip |>
  cast_dfm(
    document = doc_id,
    term = lemma,
    value = n) |>
  dfm_trim(min_docfreq = 0.1,
           max_docfreq = 0.9,
           docfreq_type = "prop")
```

```r
# Question n°1 & n°2
# Code copy-pasted from the exercise's page
k_topic <- seq(2, 20, by = 2)
opts <- list(seed = 67)

lda_2to20 <- lapply(k_topic, function(k) LDA(un.dtm, k, control = opts))
diag_list <- lapply(lda_2to20, topic_diagnostics, un.dtm, 10)
names(diag_list) <- k_topic
df_diag <- bind_rows(diag_list, .id = "K")

df_diag |>
  group_by(K) |>
  summarize(coherence = mean(topic_coherence),
            exclusivity = mean(topic_exclusivity)) |>
  ggplot(aes(x = coherence,
             y = exclusivity)) +
  geom_text(aes(label = K)) +
  labs(x = "Semantic Coherence",
       y = "Exclusivity")
```



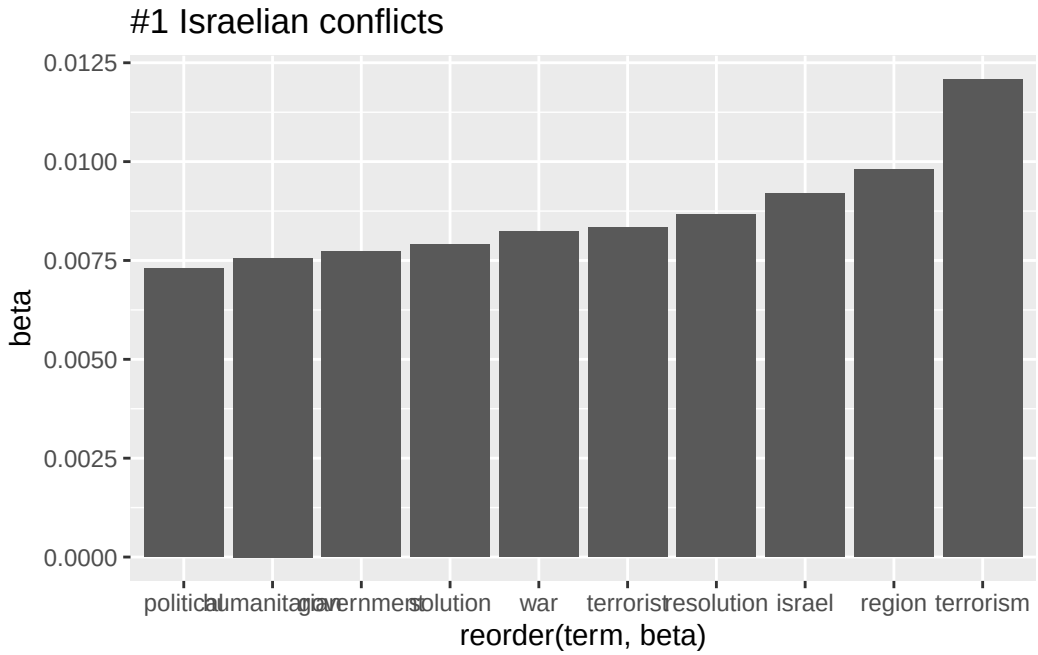
It looks like the 6 topics model is the one maximising both semantic coherence (~-22.5) & exclusivity (~9.23). Even though one can argue that there are multiple topic models with higher exclusivity (6 in total), only 2 models have close or higher semantic coherence (4 & 2).

```r
# Looking at the 6 topic models 10 top terms
opts <- list(seed = 67)
lda_6topics <- LDA(un.dtm, k = 6, control = opts)
terms(lda_6topics, 10)
```

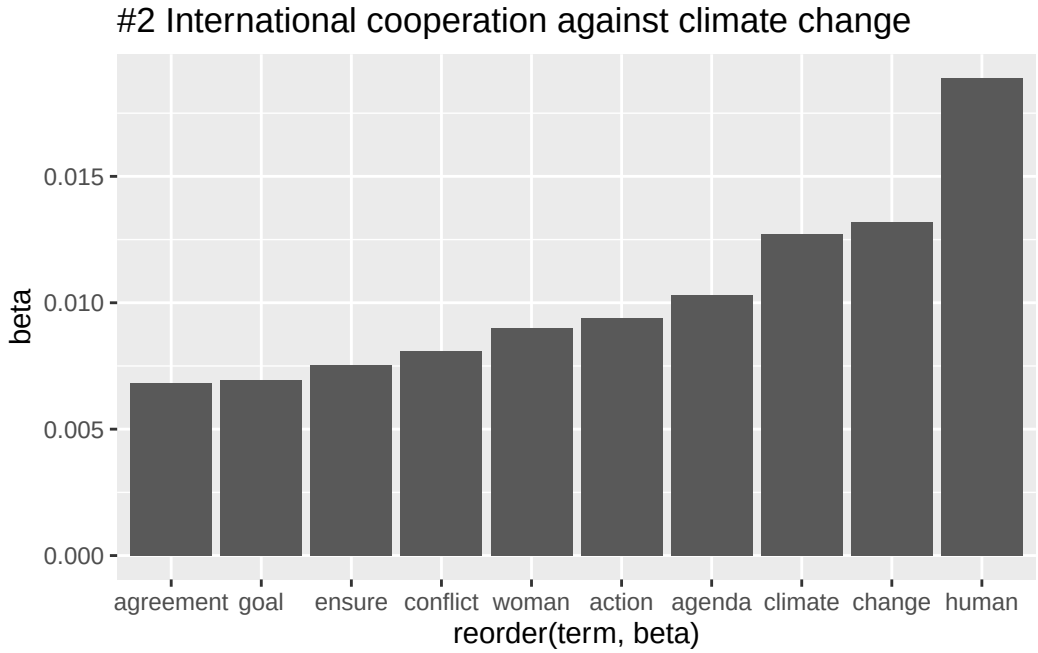
```
      Topic 1        Topic 2     Topic 3    Topic 4      Topic 5
 [1,] "terrorism"    "human"     "nuclear"  "climate"    "economic"
 [2,] "region"       "change"    "korea"    "government" "political"
 [3,] "israel"       "climate"   "north"    "island"     "cooperation"
 [4,] "resolution"   "agenda"    "time"     "change"     "goal"
 [5,] "terrorist"    "action"    "weapon"   "caribbean"  "regional"
 [6,] "war"          "woman"     "conflict" "community"  "national"
 [7,] "solution"     "conflict"  "war"      "life"       "achieve"
 [8,] "government"   "ensure"    "future"   "economic"   "stability"
 [9,] "humanitarian" "goal"      "council"  "hurricane"  "social"
[10,] "political"    "agreement" "change"   "planet"     "region"
      Topic 6
 [1,] "african"
 [2,] "council"
 [3,] "session"
 [4,] "republic"
 [5,] "agenda"
 [6,] "regard"
 [7,] "commitment"
 [8,] "community"
 [9,] "reform"
[10,] "union"
```

```r
# Question n°3 (Making a barplot)
# Preparing the variable first
library(reshape2)
topic_plot <- tidy(lda_6topics, matrix = "beta") |>
  group_by(topic) |>
  slice_max(beta, n = 10) |>
  arrange(topic, desc(beta))
```

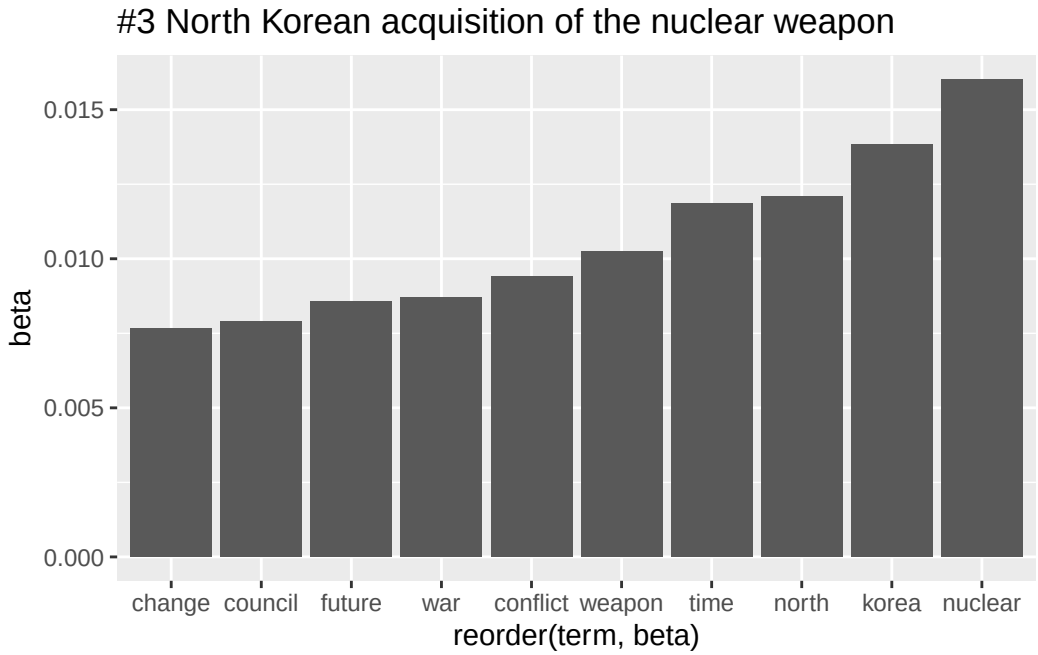
```r
# Plotting each plot at a time
# Not most elegant solution, but minimizes potential issues
ggplot(filter(topic_plot, topic == 1), aes(x=reorder(term,beta), y=beta)) + geom_col() + ggt
```



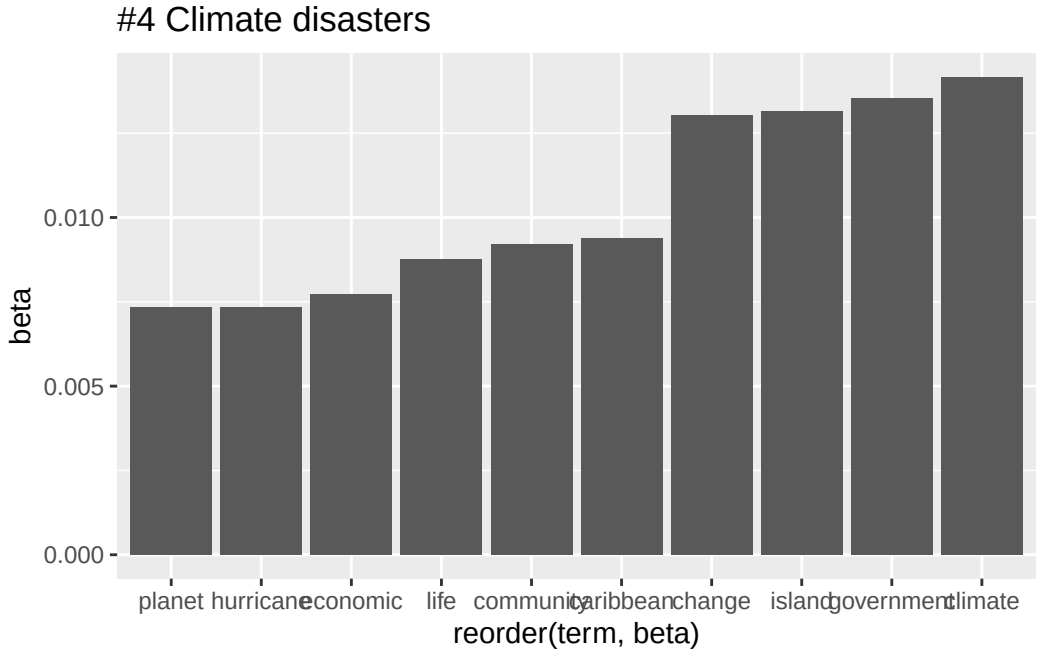

```
ggplot(filter(topic_plot, topic == 2), aes(x=reorder(term,beta), y=beta)) + geom_col() + ggti
```



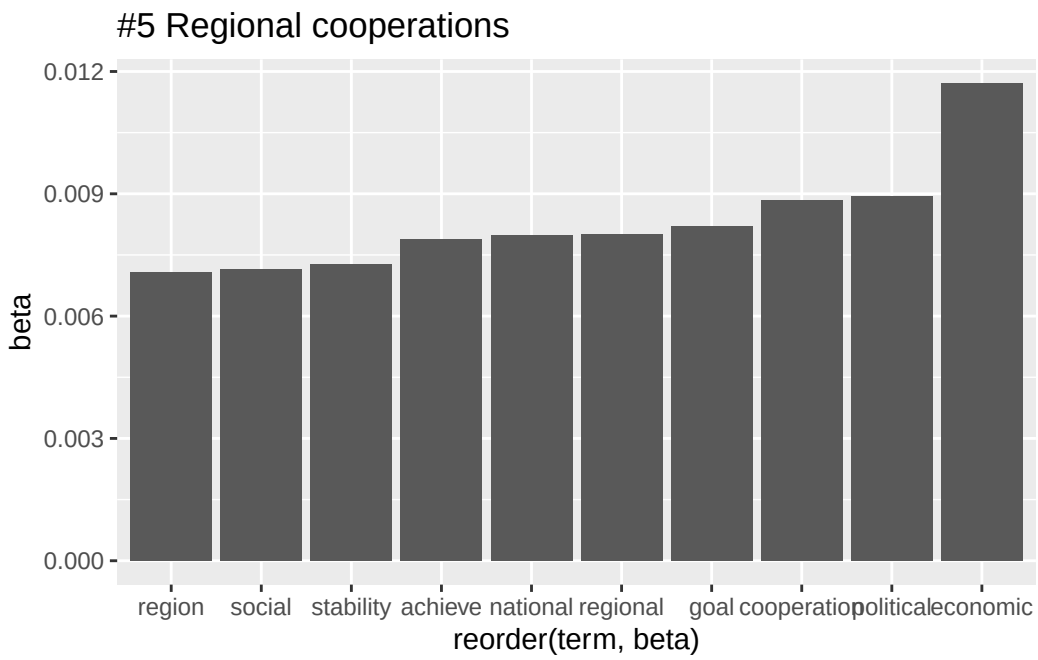
```
ggplot(filter(topic_plot, topic == 3), aes(x=reorder(term,beta), y=beta)) + geom_col() + ggti
```




```
ggplot(filter(topic_plot, topic == 4), aes(x=reorder(term,beta), y=beta)) + geom_col() + ggti
```

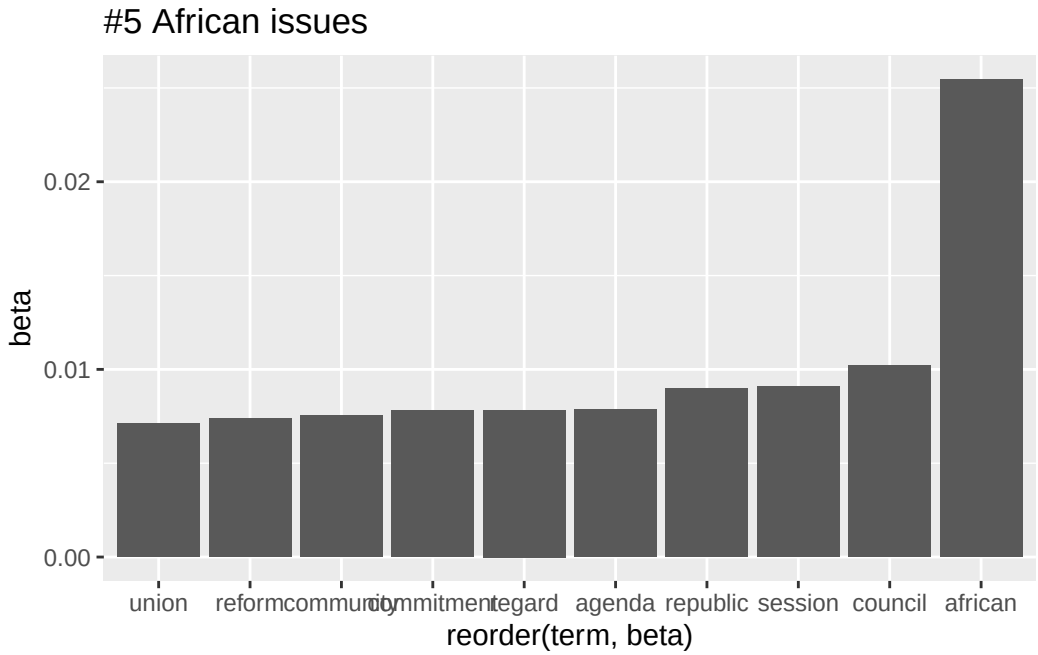


```
ggplot(filter(topic_plot, topic == 5), aes(x=reorder(term,beta), y=beta)) + geom_col() + ggti
```



The word Africa appears significantly much higher than other terms, suggesting it being the center of the topic

```
ggplot(filter(topic_plot, topic == 6), aes(x=reorder(term,beta), y=beta)) + geom_col() + ggti
```



Question n°4: the in class exercise analyzed which topics were more likely to get published in prestigious papers. Here, downstream analysis could be looking for each countries likelihood to discuss a topic such as #1 Israelian conflitcs one over time. This would require us to retrieve older and more recent statements, allowing for time-series analysis of the issue. But this is just one example of many possibilites"

Structural topic modeling

```
# Code copy-pasted from exercise page
rownames(un) <- paste0("doc", 1:nrow(un))
un <- un[rownames(un.dtm),]
```

```
# Code retrieved from "Mapping texts"
set.seed(67)
# this might take a while!
stm_env <- stm(documents = un.dtm, K =6, data = un, prevalence = ~gdp_per_capita + continent
# Thanks to onyxia, this actually did not take that long
```

```
labelTopics(stm_env)
```

```
Topic 1 Top Words:
     Highest Prob: nuclear, korea, weapon, conflict, council, north, human
     FREX: korea, north, nuclear, weapon, korean, ukraine, treaty
     Lift: north, korea, ukraine, missile, korean, nuclear, disarmament
     Score: north, korea, ukraine, nuclear, korean, weapon, missile
Topic 2 Top Words:
     Highest Prob: israel, resolution, region, terrorism, palestinian, government, humanitari
     FREX: israel, palestinian, arab, iran, occupation, palestine, territory
     Lift: israel, jerusalem, arab, occupation, muslim, palestinian, iran
     Score: israel, arab, iran, palestinian, jerusalem, occupation, yeman
Topic 3 Top Words:
     Highest Prob: government, human, life, poverty, woman, economic, cent
     FREX: poverty, cent, drug, education, democracy, woman, child
     Lift: divide, happy, drug, love, feel, wealth, corruption
     Score: divide, government, poverty, quality, drug, happy, democracy
Topic 4 Top Words:
     Highest Prob: cooperation, european, terrorism, future, time, region, political
     FREX: european, europe, migration, common, civilization, multilateralism, china
     Lift: civilization, european, europe, strick, contrary, perspective, answer
     Score: civilization, european, europe, syrian, central, minority, ethnic
Topic 5 Top Words:
     Highest Prob: african, agenda, republic, national, implementation, commitment, session
     FREX: african, implementation, regard, session, central, union, agenda
     Lift: sudan, malus, sahel, sahara, ababa, african, africa
     Score: sudan, african, malus, sahel, central, libya, union
Topic 6 Top Words:
     Highest Prob: climate, change, island, caribbean, ocean, hurricane, pacific
     FREX: island, pacific, ocean, caribbean, hurricane, climate, disaster
     Lift: pacific, irma, maria, island, ocean, fiji, hurricane
     Score: pacific, island, ocean, irma, caribbean, hurricane, climate
```

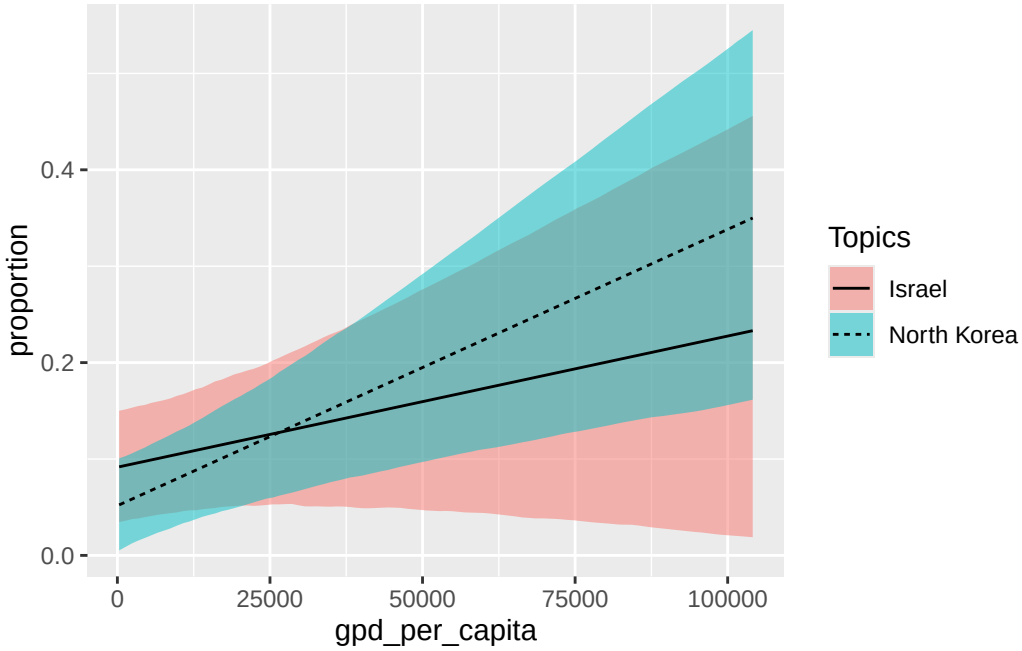
```
stm_effect <- estimateEffect(c(1, 2) ~gdp_per_capita + continent, stm_env, metadata=un)
```

```
# The next step requires stminsights package
install.packages("stminsights")
```

```
Installing package into '/usr/local/lib/R/site-library'
(as 'lib' is unspecified)
```

```
library(stminsights)
stm_effect <- get_effects(stm_effect, variable ="gdp_per_capita", type ="continuous")|>
```

```
  mutate(Topics = case_when(topic==1~"North Korea",topic==2~"Israel"))
# Plotting the effects
stm_effect|>
  ggplot(aes(x =value,y =proportion,fill =Topics))+geom_ribbon(aes(ymin =lower,ymax =upper),a
```



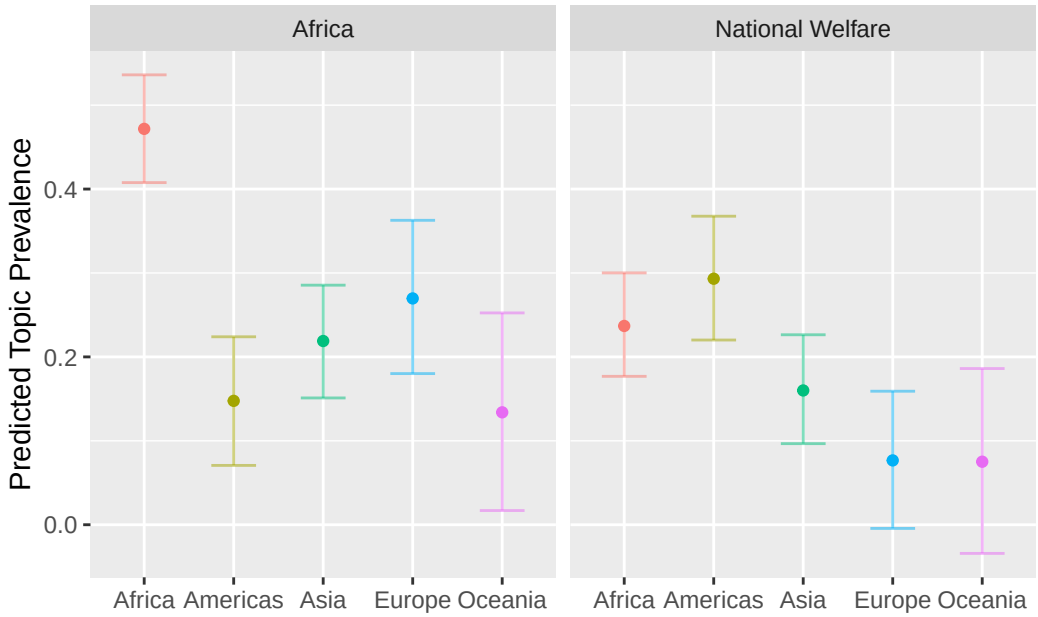
Both topic prevalence show a positive relationship with gdp per capita, with a stronger relationship in North Korea's case. In substantive terms, the more a country's gdp is high, the more Israel et North Korea topics are prevalent in its 2017 UN General Assembly speech. It would be interesting to find a topic that does not follow this trend and comparee the results.

```
#Restarting the analysis
stm_effect <- estimateEffect( ~gdp_per_capita + continent, stm_env, metadata=un)

# Code copy-pasted from the class website & adapted to local object names
stm_effect <- get_effects(stm_effect, variable = "continent",
                          type = "pointestimate") |>
  mutate(Topics = case_when(
    topic == 5 ~ "Africa",
    topic == 3 ~ "National Welfare")
  )

# NAs had to be filtered
stm_effect |>
```

```
  filter(!is.na(Topics)) |>
  ggplot(aes(x = value, y = proportion, color = value)) +
  geom_errorbar(aes(ymin = lower, ymax = upper), alpha = 0.5, width = .5) +
  geom_point() +
  labs(x = "") +
  guides(color = "none") +
  labs(y = "Predicted Topic Prevalence") +
  facet_wrap(~Topics)
```



Question n°4: the topic of Africa is significantly more prevalent in African speeches (over 0.4 compared to ~0.2 to ~0.375 for Europe, which has closest prevalence), which means African countries tend to discuss more about their own issues. By progressive order of magnitude, Africa is discussed less by Oceanian countries (~0 to ~0.25), then Americas (~0.075 to ~0.21), Asia (~0.15 to ~0.3) and then Europe (~0.2 to ~0.375). As for the topic of National Welfare, it appears as significantly more prevalent in Africa (~0.18 to ~0.3) and Americas (~0.2 to ~0.38) speeches than Europe (0 to ~0.18) and Oceania (negative to ~0.18). The plot still shows that Asian speeches share National Welfare prevalence with Africa and Americas, going from 0.1 to ~0.22. We could infer from this that European & Oceania statements do not focus as much on national welfare compared to Africa & Americas.

On your own

```
# Question n°1: lets check the 6 topics again through a STM, and then select & label one
# Copy-pasted & modified from "Mapping texts"
stm_oyo <- tidy(stm_env, matrix ="beta")|>
  group_by(topic)|>
  slice_max(beta,n =6,with_ties =FALSE)|>

summarize(terms =paste0(term,collapse =", "))
print(stm_oyo,n =10)
```

```
# A tibble: 6 x 2
  topic terms
  <int> <chr>
1     1 nuclear, korea, weapon, conflict, council, north
2     2 israel, resolution, region, terrorism, palestinian, government
3     3 government, human, life, poverty, woman, economic
4     4 cooperation, european, terrorism, future, time, region
5     5 african, agenda, republic, national, implementation, commitment
6     6 climate, change, island, caribbean, ocean, hurricane
```

The topic previously labelled as climate disasters looks like an interesting one (n°6), lets use it for the next questions (question no2)

```
# Question n°2: let's use findThoughts() to see the top 3 statements related to topic 6
# Line truncation options have been removed to read the entire statements
corpus_top3 <- findThoughts(stm_env, texts = un$text.clean, topics = 6, n = 3)
corpus_top3$docs
```

```
$`Topic 6`
[1] "i want to begin by congratulating mr. miroslav lajcak on assuming the presidency of the
hosted with sweden in june was an unqualified success. as a global community, we have begun
case scenario in which their homes sink beneath the waves. for the fijian people, climate cha
destructive for the world and for each nation. it may be tempting for political leaders to sl
being of the nations and ecosystems of the world. if we view the process as some sort of nego
cop meeting. we are already delighted to see the energy and sense of purpose of the leaders
energy solutions for countries across the world, which offers great promise that we can achie
going canoe - a na drua - in the main hall in bonn to remind everyone of the need to fill its
```

```
[2] "i come to the general assembly straight from the front line of the war on climate change
atlantic and line up to pound us with maximum force and fury. prior to this century, no other
first session of conference of the parties to the united nations framework convention on clim
induced global warming. let it spark a thousand points of light, not shame. let us take serio
[3] "i am honoured to address the general assembly on behalf of the people and the government
```

After further reading, the coherence between pre-established label and actual content of the statement can be confirmed. However, multiple observations emerge from an in depth qualitative reading. First, we discover that all statement were made by island countries, and even two carribean countries (Dominica & Dominican Republic, Fiji on the other hand being in the Pacific). Quantitative analysis did include the word carribeans in its top 10 prevalent words, but was not considered much due to the ambiguity of whether it was related to a specific event or to the statement's authorship. This clears a previous ambiguity. All countries do speak about climate change indeed, and we can now infer that the carribeans were so prevalent due to the Irma hurricane that had just happened before. However, each statement contains notable nuances. Dominica specifically points the responsibility of richer countries in global carbon emissions & the following climate disasters: "we are shouldering the consequences of the actions of others, actions that endanger our very existence and all for the enrichment of a few elsewhere", the tone is also more martial "i come to the general assembly straight from the front line of the war on climate change". Dominican Republic (statement 3) conveys a more cooperative speech, highlighting the importance of solidarity: "we are facing a growing problem that knows no borders and requires the cooperation and solidarity of all of us". Quantitative analysis managed to detect the topic with high accuracy, but the specific metrics used here did not account for potential nuance in tone & rhetoric of each statement. Another point of interest is the inclusion of economic solutions brought by actors: "the world bank facility must be recapitalized for greater coverage by using already committed funds from the green climate fund", "my country is once again proposing to the assembly that we create a special fund for dealing with this type of disaster". This was not necessarily captured by our model. All these results reinforce our model's predictions of topics & other quantitative analysis, but a more qualitative study of the corpus brought much more context and nuance on the statements, showing that NLP analysis cannot reside in a single model to answer all potential questions, and that multiple methods must be considered during the research's design for text analysis.